

G1_Task

Hello,

Please read the following carefully:

As a Software Engineering research team at the Polytechnique Montréal (Canada), we are interested in understanding developers' assessment of coding task difficulty. You will be provided with a series of 12 coding questions. Questions are divided into two parts:

- First, you will be given three instructions representing coding tasks and you will be asked to write, in a few words, what is the underlying task represented by all these instructions. The instructions represent the same tasks, but they vary in wording and information they give. In a nutshell, you have to distill the essence of the task and write it in a few words.

- In a second time, you will be asked to assess how difficult implementing this **task** would be. Note we ask the assessment based on the task you will have worded in the previous part, not based on any of the instructions presented earlier. First, you will be asked your subjective judgment on a scale from Very Easy to Very Difficult. Then, to make judging the difficulty more relatable, we adopt a similar approach as what is used in agile development and planning poker, i.e. you will be asked to evaluate how long it would take to implement a task (in minutes here). Consider in your assessment the knowledge needed to implement the task, the external information you would need to use, whether you would need the assistance of a LLM such as ChatGPT (and so the time it would take to debug the obtained code) etc. Finally, you will have to explain in a few words why you gave such difficulty ratings.

To give you an idea of the range of tasks you will need to assess, the first section of this questionnaire will be two examples we devised to illustrate the questionnaire.

Completing the task will take about 30 minutes.

Note also that all examples are written using Python 3.10 syntax.

Please make sure to answer all the questions to the best of your ability.

Please note that your identity and personal data will not be disclosed, while we plan to use the aggregated results and anonymized responses as part of a scientific publication. If you have any questions about the questionnaire or our research, please do not hesitate to contact us.

By proceeding, you consent to take part in the study under the conditions described previously.

Thank you,

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* Indique une question obligatoire

1. Please enter your prolific ID : *

Example of tasks

As we ask of you some estimation of difficulty and the tasks dealt in this questionnaire can be a bit different from what you can be used to as a developer, we give you examples of some tasks along with some assessments.

Note that this assessment is **our assessment**. You may have a different opinion based on your background. The idea of those examples is to give an extreme range of what type of tasks you can expect in the questionnaire.

As a reminder: You will first be given three instructions and asked to explain what the task is about in a few words. Then, you will have to rate the difficulty of the **task** you just described (not accounting for particular instruction). Rating the difficulty is done in two steps: subjective difficulty and approximate time needed. Finally, you will be asked to say why you gave such rating.

Example 1

Instruction 1

```
1 def subtract(c1, c2):  
2     """  
3     Deliver the result of subtracting two complex numbers,  
4     specified as 'c1' and 'c2', as a complex output.  
5     :param c1: The first complex number, complex.  
6     :param c2: The second complex number, complex.  
7     :return: The difference of the two complex numbers, complex.  
8     """
```

Instruction 2

```
1 def subtract(c1, c2):
2     """
3     The method receives two complex numbers 'c1' and 'c2', from which it
4     subtracts 'c2.real' from 'c1.real' to determine the real portion, and
5     subtracts 'c2.imag' from 'c1.imag' to determine the imaginary portion.
6     It constructs and returns a new complex number using these results
7     with the 'complex()' function.
8     :param c1: The first complex number,complex.
9     :param c2: The second complex number,complex.
10    :return: The difference of the two complex numbers,complex.
11    """
```

Instruction 3

```
1 def subtract(c1, c2):
2     """
3     Subtracts two complex numbers.
4     :param c1: The first complex number,complex.
5     :param c2: The second complex number,complex.
6     :return: The difference of the two complex numbers,complex.
7     >>> complexCalculator = ComplexCalculator()
8     >>> complexCalculator.subtract(1+2j, 3+4j)
9     (-2-2j)
10    """
```

Answer

The task encapsulated by the **subtract**

function

is to compute the subtraction between two complex numbers.

This task has a subjective difficulty of **Very Easy** as assessed by us and would take about **1 min**

to implement. There is no particular difficulty, the task is very straightforward without any corner case, it just requires to know what a complex is, and the standard Python library for it.

Example 2

Context

```
1 import numpy as np
2 class MetricsCalculator2:
3     """
4     The class provides to calculate Mean Reciprocal Rank (MRR) and Mean Average Precision (MAP)
5     based on input data, where MRR measures the ranking quality and MAP measures the average precision.
6     """
7
8     def __init__(self):
9         pass
10
11     @staticmethod
12     def map(data):
13         pass
```

Instruction 1

```
1 def mrr(data):
2     """
3     Compute the MRR of the input data. MRR is a widely used evaluation index.
4     It is the mean of reciprocal rank.
5     :param data: the data must be a tuple, list 0,1, eg. ([1,0,...],5).
6     In each tuple (actual result,ground truth num),ground truth num is the total ground num.
7     ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5),([1,0,...],6),([0,0,...],5)].
8     1 stands for a correct answer, 0 stands for a wrong answer.
9     :return: if input data is list, return the recall of this list. if the input data is list of list, return the
10     average recall on all list. The second return value is a list of precision for each input.
11     >>> MetricsCalculator2.mrr([([1, 0, 1, 0], 4)])
12     >>> MetricsCalculator2.mrr([([1, 0, 1, 0], 4), ([0, 1, 0, 1], 4)])
13     1.0, [1.0]
14     0.75, [1.0, 0.5]
15
16     """
```

Instruction 2

```
1 def mrr(data):
2     """
3     Assess the Mean Reciprocal Rank (MRR) from the input 'data'.
4     MRR assesses the average of reciprocal ranks of outcomes.
5     The input structure 'data' must be a tuple consisting of a list of
6     binary values and its total count or a list of such tuples.
7     Each binary value (0 or 1) represents the correctness of an answer.
8     If 'data' is a tuple, the function returns its mean reciprocal rank.
9     If 'data' is a list, the function returns the overall average MRR along
10     with a list showing reciprocal ranks for each component tuple.
11
12     :param data: the data must be a tuple, list 0,1, eg. ([1,0,...],5).
13     In each tuple (actual result,ground truth num),ground truth num is the total ground num.
14     ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5),([1,0,...],6),([0,0,...],5)].
15     1 stands for a correct answer, 0 stands for a wrong answer.
16     :return: if input data is list, return the recall of this list. if the input data is list of list, return the
17     average recall on all list. The second return value is a list of precision for each input.
18
19     """
```

Instruction 3

```
1 def mrr(data):
2     """
3     Evaluate the Mean Reciprocal Rank (MRR) from 'data'. This metric calculates the mean of reciprocal
4     ranks for correct answers. Input 'data' could be either a tuple containing binary correctness indicators
5     and the aggregate count of answers, or multiple tuples of this type in a list. For each tuple, the function
6     measures reciprocal ranks by multiplying the binary indicators by reciprocals of their positions, finding the
7     earliest nonzero value to establish the rank of the first correct response. If the input 'data' is a tuple,
8     the MRR calculated for this specific tuple is returned; if it's a list, the function summarizes by providing
9     both the average MRR and a list of individual MRR values for each tuple.
10
11     :param data: the data must be a tuple, list 0,1, eg. ([1,0,...],5).
12     In each tuple (actual result, ground truth num), ground truth num is the total ground num.
13     ([1,0,...],5), or list of tuple eg. [([1,0,1,...],5), ([1,0,...],6), ([0,0,...],5)].
14     1 stands for a correct answer, 0 stands for a wrong answer.
15     :return: if input data is list, return the recall of this list. if the input data is list of list, return the
16     average recall on all list. The second return value is a list of precision for each input.
17
18     """
```

Answer

The task encapsulated by the **mrr** function is to compute the Mean Reciprocal Rank of binary tuples data.

This task has a subjective difficulty of **Very Hard** as assessed by us and would take about **40 min**

to implement. The difficulty mainly branches from dealing with the tuple data and how the formula is processed accordingly (with a few corner cases). This can take additional time if the Mean Reciprocal Rank is not known.

Question 1:

Below are instructions for implementing the function **x_or_y**.

Instruction 1

```
1 def x_or_y(n, x, y):
2     """A simple program which should return the value of x if n is
3     a prime number and should return the value of y otherwise.
4
5     Examples:
6     for x_or_y(7, 34, 12) == 34
7     for x_or_y(15, 8, 5) == 5
8
9     """
```

Instruction 2

```
1 def x_or_y(n, x, y):  
2     """  
3     Construct a function named 'x_or_y' with three parameters:  
4     an integer 'n', and two variables 'x' and 'y'. This function embeds  
5     a helper function 'is_prime(a)' that assesses if 'n' is a prime.  
6     'is_prime()' returns 'True' when 'n' is more than 1 and does not  
7     have divisors other than 1 and itself (up to the square root of 'n'),  
8     and 'False' otherwise. The overall function delivers 'x' if 'is_prime(n)'  
9     concludes 'True', otherwise 'y'.  
10    """
```

Instruction 3

```
1 def x_or_y(n, x, y):  
2     """  
3     Create a function titled 'x_or_y' accepting three arguments: an integer 'n',  
4     and two variables 'x' and 'y'. Inside, utilize a nested helper function that  
5     verifies prime status of 'n' by testing divisibility other than by 1 or itself.  
6     Depending on the result, return 'x' if 'n' is prime, otherwise return 'y'.  
7     """
```

2. In a few words, what is the task this function aims to implement? *

3. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

4. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

5. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 2:

Below are instructions for implementing the function:

Context

```
1 import random
2 class Snake:
3     """
4     The class is a snake game, with allows snake to move and eat food,
5     and also enables to reset, and generat a random food position.
6     """
7
8     def __init__(self, SCREEN_WIDTH, SCREEN_HEIGHT, BLOCK_SIZE, food_position):
9         """
10         Initialize the length of the snake, screen width, screen height,
11         block size, snake head position, score, and food position.
12         :param SCREEN_WIDTH: int
13         :param SCREEN_HEIGHT: int
14         :param BLOCK_SIZE: int, Size of moving units
15         :param food_position: tuple, representing the position(x, y) of food.
16         """
17         self.length = 1
18         self.SCREEN_WIDTH = SCREEN_WIDTH
19         self.SCREEN_HEIGHT = SCREEN_HEIGHT
20         self.BLOCK_SIZE = BLOCK_SIZE
21         self.positions = [((SCREEN_WIDTH / 2), (SCREEN_HEIGHT / 2))]
22         self.score = 0
23         self.food_position = food_position
24
25     def move(self, direction):
26         pass
27
28     def random_food_position(self):
29         pass
30
31     def reset(self):
32         pass
33
```

Instruction 1

```
1 def eat_food(self):
2     """
3     Increase the length of the snake by 1 and increase the score by 100.
4     Randomly generate a new food position, but don't place it on the snake.
5     :return: None
6     >>> snake = Snake(100, 100, 1, (51, 51))
7     >>> snake.move((1,1))
8     >>> snake.eat_food()
9     self.length = 2
10    self.score = 10
11
12    """
```


Instruction 2

```
1 def eat_food(self):
2     """
3     Augment the snake's overall length by one increment,
4     boost the score by 100 points, and call a method to randomly deploy
5     a new food item without overlapping the snake's segments.
6     This function aims to simulate the feeding and subsequent growth of the snake.
7     Concurrently, it provides reward points to reflect the food eaten.
8     After enhancing the size and score, a specific method is triggered to ensure
9     new food appears on the board where the snake is not present.
10    :return: None
11    """
```

Instruction 3

```
1 def eat_food(self):
2     """
3     Enhance the snake's body length by 1 unit, add 100 to the player's score,
4     and execute a method that will position new food at a random location away
5     from the snake. The goal is to mimic the snake's growth after consumption
6     and reward the player with extra points. Following these updates,
7     the function initiates a separate method to set up the new food item on
8     the playing field, ensuring it does not coincide with the snake's current location.
9     :return: None
10    """
```

6. In a few words, what is the task this function aims to implement? *

7. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

☐ Very Easy

☐ Easy

☐ Not Easy Nor Difficult

☐ Difficult

☐ Very Difficult

8. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

9. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 3:

Below are instructions for implementing the function:

Instruction 1

```
1 def string_to_md5(text):
2     """
3     Given a string 'text', return its md5 hash equivalent string.
4     If 'text' is an empty string, return None.
5
6     >>> string_to_md5('Hello world') == '3e25960a79dbc69b674cd4ec67a72c62'
7     """
```

Instruction 2

```
1 def string_to_md5(text):
2     """
3     Define a function called 'string_to_md5' which receives 'text' as an argument.
4     If 'text' is empty, the function needs to return None. For any non-empty string,
5     the function should use the 'hashlib' for computing an md5 hash. This includes
6     creating a new md5 object 'm' with 'hashlib.md5()', updating this object with
7     'text' encoded to 'utf-8' in the form of 'm.update(text.encode('utf-8'))', and lastly,
8     the function yields the hexadecimal version of the hash with 'm.hexdigest()'.
9     """
```

Instruction 3

```
1 def string_to_md5(text):
2     """
3     Write a function named "string_to_md5" that takes a string "text" as its input.
4     If "text" is an empty string, the function returns None. For non-empty strings,
5     the function should utilize the "hashlib" module to compute the md5 hash.
6     It uses "hashlib.md5()" to create a new md5 hash object "m".
7     Then, it updates the hash object "m" with "text" encoded in "utf-8" using
8     "m.update(text.encode("utf-8"))". Finally, the function returns the hexadecimal
9     equivalent of the hash using "m.hexdigest()".
10    """
```

10. In a few words, what is the task this function aims to implement? *

11. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

☐ Very Easy

☐ Easy

☐ Not Easy Nor Difficult

☐ Difficult

☐ Very Difficult

12. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

13. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 4:

Below are instructions for implementing the function:

Context

```
1 class Order:
2     """
3     The class manages restaurant orders by allowing the addition of dishes,
4     calculation of the total cost, and checkout.
5     """
6
7     def __init__(self):
8         """
9         Initialize the order management system
10        self.menu stores the dishes of restaurant inventory
11        menu = [{"dish": dish name, "price": price, "count": count}, ...]
12        self.selected_dishes stores the dishes selected by customer
13        selected_dish = {"dish": dish name, "count": count, "price": price}
14        self.sales stores the sales of each dish
15        sales = {dish name: sales}
16        """
17        self.menu = []
18        self.selected_dishes = []
19        self.sales = {}
20
21
22    def add_dish(self, dish):
23        pass
24
25    def calculate_total(self):
26        pass
```

Instruction 1

```
1 def checkout(self):
2     """
3     Check out the dishes ordered. IF the self.selected_dishes is not empty,
4     invoke the calculate_total method to check out.
5     :return False if the self.selected_dishes is empty,
6     or total(return value of calculate_total) otherwise.
7
8     >>> order = Order()
9     >>> order.menu.append({"dish": "dish1", "price": 10, "count": 5})
10    >>> order.sales = {"dish1": 0.8}
11    >>> order.add_dish({"dish": "dish1", "price": 10, "count": 4})
12    True
13    >>> order.checkout()
14    32.0
15    """
```

Instruction 2

```
1 def checkout(self):
2     """
3     Handle the checkout for the orders listed in 'self.selected_dishes'.
4     If dishes exist in this list, apply the 'calculate_total' function
5     to calculate the total price and return it. If the list is devoid
6     of dishes, return 'False'; if not, provide the total calculated by
7     'calculate_total'.
8     :return False if the self.selected_dishes is empty, or total(return
9     value of calculate_total) otherwise.
10
11     """
```

Instruction 3

```
1 def checkout(self):
2     """
3     Process the checkout for the ordered dishes. If 'self.selected_dishes'
4     contains items, use 'calculate_total' to determine and return the total
5     price of the ordered dishes considering their counts and costs.
6     After determining the total, clear 'self.selected_dishes'.
7     Return 'False' if 'self.selected_dishes' contains no items; if not,
8     deliver the calculated total reflecting the aggregate cost of all the dishes chosen.
9     :return False if the self.selected_dishes is empty, or total(return value
10     of calculate_total) otherwise.
11     """
```

14. In a few words, what is the task this function aims to implement? *

15. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

☐ Very Easy

☐ Easy

☐ Not Easy Nor Difficult

☐ Difficult

☐ Very Difficult

16. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

17. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 5:

Below are instructions for implementing the function:

Context

```
1 import math
2
3
4 def poly(xs: list, x: float):
5     """
6     Evaluates polynomial with coefficients xs at point x.
7     return xs[0] + xs[1] * x + xs[1] * x^2 + .... xs[n] * x^n
8     """
9     return sum([coeff * math.pow(x, i) for i, coeff in enumerate(xs)])
```

Instruction 1

```
1 def find_zero(xs: list):
2     """ xs are coefficients of a polynomial.
3     find_zero find x such that poly(x) = 0.
4     find_zero returns only one zero point, even if there are many.
5     Moreover, find_zero only takes list xs having even number of coefficients
6     and largest non zero coefficient as it guarantees
7     a solution.
8     >>> round(find_zero([1, 2]), 2) # f(x) = 1 + 2x
9     -0.5
10    >>> round(find_zero([-6, 11, -6, 1]), 2) # (x - 1) * (x - 2) * (x - 3) = -6 + 11x - 6x^2 + x^3
11    1.0
12    """
```

Instruction 2

```
1 def find_zero(xs: list):
2     """
3     Formulate a function called 'find_zero' utilizing a list 'xs' that
4     contains polynomial coefficients, to identify a polynomial root through
5     the Newton-Raphson method. It begins with an assumed root 'x' at zero,
6     refining this estimate by sequentially evaluating the polynomial and its
7     derivative at 'x' with functions 'poly' and 'derivative', using list
8     comprehensions for computations. If the polynomial value exceeds a minimal
9     tolerance 'tol' (specified at 1e-5), it updates 'x' by employing 'x - fx / dfx',
10    where 'fx' is the current polynomial value at 'x' and 'dfx' is its derivative.
11    The refinement cycle continues for up to 1000 iterations or until the polynomial
12    is below the tolerance level. The functionality is confined to polynomials that
13    possess an even number of coefficients and a maximum non-zero coefficient that
14    indicates a root can be found.
15    """
```

Instruction 3

```
1 def find_zero(xs: list):
2     """
3     Construct a function named 'find_zero' which requires a list 'xs'
4     representing coefficients of a polynomial. The function seeks to locate
5     a root of the polynomial through the Newton-Raphson iterative method.
6     It starts with an initial estimate 'x' of 0, and repeatedly narrows down
7     on a root by computing the polynomial and its derivative at 'x' with the
8     'poly' and 'derivative' functions respectively, using list comprehensions.
9     It checks if the computed polynomial value is below a small tolerance 'tol'
10    (set to 1e-5). If above this value, it recalculates 'x' by applying the
11    formula 'x - fx / dfx', where 'fx' is the polynomial's value at 'x'
12    and 'dfx' is its derivative. The iteration stops after at most 1000
13    cycles or once the target tolerance level is reached. This function only
14    functions accurately for polynomials with even coefficients and where the
15    topmost non-zero coefficient guarantees a root can be found.
16    """
```

18. In a few words, what is the task this function aims to implement? *

19. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

20. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

21. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 6:

Below are instructions for implementing the function:

Context

```
1 from datetime import datetime, timedelta
2 class CalendarUtil:
3     """
4     This is a class as CalendarUtil that provides functionalities to manage calendar events,
5     schedule appointments, and perform conflict checks.
6     """
7
8     def __init__(self):
9         """
10         Initialize the calendar with an empty list of events. Each event added is a dictionary.
11         """
12         self.events = []
13
14     def add_event(self, event):
15         pass
16
17     def remove_event(self, event):
18         pass
19
20     def get_events(self, date):
21         pass
22
23     def is_available(self, start_time, end_time):
24         pass
25
26     def get_available_slots(self, date):
27         pass
```

Instruction 1

```
1 def get_upcoming_events(self, num_events):
2     """
3     Get the next n upcoming events from a given date.
4     :param n: The number of upcoming events to get,int.
5     :return: A list of the next n upcoming events from the given date,list.
6     >>> calendar = CalendarUtil()
7     >>> calendar.events = [{'date': datetime(2025, 1, 1, 0, 0),
8     'start_time': datetime(2025, 1, 1, 0, 0), 'end_time': datetime(2025, 1, 1, 23, 0),
9     'description': 'New Year'},{'date': datetime(2025, 1, 2, 0, 0),'end_time': datetime(2025, 1, 2, 1, 0),
10    'description': 'New Year 2'}]
11    >>> calendar.get_upcoming_events(1)
12    [{'date': datetime.datetime(2025, 1, 1, 0, 0), 'start_time': datetime.datetime(2025, 1, 1, 0, 0),
13    'end_time': datetime.datetime(2025, 1, 1, 23, 0), 'description': 'New Year'},
14    {'date': datetime.datetime(2025, 1, 2, 0, 0), 'end_time': datetime.datetime(2025, 1, 2, 1, 0),
15    'description': 'New Year 2'}]
16    """
```

Instruction 2

```
1 def get_upcoming_events(self, num_events):
2     """
3     Get the next "n" upcoming events starting today's date.
4     The function should determine events that commence on
5     or after today and return up to "n" of these events.
6     :param n: The number of upcoming events to get,int.
7     :return: A list of the next n upcoming events from the given date,list.
8     """
```

Instruction 3

```
1 def get_upcoming_events(self, num_events):  
2     """  
3     Procure the successive 'n' upcoming events from today's date.  
4     The method should recognize events that start from today onwards  
5     and return a maximum of 'n' such events.  
6     :param n: The number of upcoming events to get,int.  
7     :return: A list of the next n upcoming events from the given date,list.  
8     """
```

22. In a few words, what is the task this function aims to implement? *

23. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

24. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

25. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 7:

Below are instructions for implementing the function:

Instruction 1

```
1 def truncate_number(number: float) -> float:
2     """
3     Develop a function named 'truncate_number' which receives a positive floating point number.
4     The goal of this function is to isolate and return the fractional part of the number.
5     This fraction can be obtained by removing the highest integer value less than or equal
6     to the number from the number itself.
7     The operation results in the digital section of the number that is present after the decimal.
8     """
```

Instruction 2

```
1 def get_upcoming_events(self, num_events):
2     """
3     Get the next "n" upcoming events starting today's date.
4     The function should determine events that commence on
5     or after today and return up to "n" of these events.
6     :param n: The number of upcoming events to get,int.
7     :return: A list of the next n upcoming events from the given date,list.
8     """
```

Instruction 3

```
1 def truncate_number(number: float) -> float:
2     """
3     Given a positive floating point number, it can be decomposed into
4     and integer part (largest integer smaller than given number) and decimals
5     (leftover part always smaller than 1).
6
7     Return the decimal part of the number.
8     >>> truncate_number(3.5)
9     0.5
10    """
```

26. In a few words, what is the task this function aims to implement? *

27. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

28. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

29. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 8:

Below are instructions for implementing the function:

Context

```
1 import re
2 class RegexUtils:
3     """
4     The class provides to match, find all occurrences, split,
5     and substitute text using regular expressions.
6     It also includes predefined patterns,
7     validating phone numbers and extracting email addresses.
8     """
9
10    def match(self, pattern, text):
11        pass
12
13    def findall(self, pattern, text):
14        pass
15
16    def split(self, pattern, text):
17        pass
18
19    def sub(self, pattern, replacement, text):
20        pass
21
22    def generate_email_pattern(self):
23        pass
24
25    def generate_phone_number_pattern(self):
26        pass
27
28    def generate_split_sentences_pattern(self):
29        pass
30
31    def validate_phone_number(self, phone_number):
32        pass
33
34    def extract_email(self, text):
35        pass
```

Instruction 1

```
1 def split_sentences(self, text):
2     """
3     Chop the specified 'text' into distinct sentences, excluding punctuation
4     from every sentence except for the ending one. The list of these sentences
5     should be returned by the function.
6     Split the text into a list of sentences without Punctuation except the
7     last sentence
8     :param text: Text to be split
9     :return: Split Text List
10    """
```

Instruction 2

```
1 def split_sentences(self, text):
2     """
3     Split the given "text" into a list of sentences, excluding punctuation
4     from all sentences except the last one. The function begins by generating
5     a pattern that identifies endings of sentences, typically including punctuation.
6     It then uses this pattern to split the text. Return the resultant list of
7     sentences, ensuring that all except the last retain their end punctuation stripped off.
8     :param text: Text to be split
9     :return: Split Text List
10    """
```

Instruction 3

```
1 def split_sentences(self, text):
2     """
3     Split the text into a list of sentences without Punctuation except the last sentence
4     :param text: Text to be split
5     :return: Split Text List
6     >>> ru = RegexUtils()
7     >>> ru.split_sentences("Aaa. Bbbb? Ccc!")
8     ['Aaa', 'Bbbb', 'Ccc!']
9     """
```

30. In a few words, what is the task this function aims to implement? *

31. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

32. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

33. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 9:

Below are instructions for implementing the function:

Instruction 1

```
1 def sort_array(arr):
2     """
3     Construct a function designated 'sort_array' to handle an array of non-negative integers.
4     It sorts the array firstly based on the count of ones in each integer's binary representation.
5     Where counts are identical, it sorts by the integer's natural decimal value.
6     Include a comparator function that sets this sorting criteria, and incorporate it into
7     a sorting algorithm that can utilize a custom comparison function.
8     """
```

Instruction 2

```
1 def sort_array(arr):
2     """
3     Create a function called 'sort_array' that takes an array of non-negative integers.
4     This function should sort the array by first evaluating the number of ones in the binary
5     form of each integer. If integers have an equal number of ones, they should be sorted
6     by their corresponding decimal values. A custom comparison function should be employed
7     to achieve this sorting, which is then utilized in a sorting algorithm accepting a custom comparator.
8     """
```

Instruction 3

```
1 def sort_array(arr):
2     """
3     In this Kata, you have to sort an array of integers according to
4     number of ones in their binary representation in ascending order.
5     For similar number of ones, sort based on decimal value.
6
7     It must be implemented like this:
8     >>> sort_array([1, 5, 2, 3, 4]) == [1, 2, 3, 4, 5]
9     >>> sort_array([-2, -3, -4, -5, -6]) == [-6, -5, -4, -3, -2]
10    >>> sort_array([1, 0, 2, 3, 4]) [0, 1, 2, 3, 4]
11    "
12
13    """
```

34. In a few words, what is the task this function aims to implement? *

35. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

☐ Very Easy

☐ Easy

☐ Not Easy Nor Difficult

☐ Difficult

☐ Very Difficult

36. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

37. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 10:

Below are instructions for implementing the function:

Context

```
1 class URLHandler:
2     """
3     The class supports to handle URLs, including extracting the scheme, host, path, query parameters, and fragment.
4     """
5
6     def __init__(self, url):
7         """
8         Initialize URLHandler's URL
9         """
10        self.url = url
11
12    def get_host(self):
13        pass
14
15    def get_path(self):
16        pass
17
18    def get_query_params(self):
19        pass
20
21    def get_fragment(self):
22        pass
23
24
25
```

Instruction 1

```
1 def get_scheme(self):
2     """
3     Fetch the scheme from 'self.url', with a successful extraction
4     resulting in the scheme returned as a string. The 'get_scheme'
5     operation identifies the end of the scheme by searching
6     for '://' in 'self.url'. If this delimiter is located, marked
7     by 'scheme_end' being other than -1, the string before 'scheme_end'
8     is returned. Should 'scheme_end' be -1, indicating no delimiter is
9     found, 'None' is returned as an indication that 'self.url' lacks a scheme.
10    :return: string, If successful, return the scheme of the URL
11    """
```

Instruction 2

```
1 def get_scheme(self):
2     """
3     Search for the position of '://' in 'self.url' to find where the scheme ends.
4     If found, return the substring from the beginning to this position as the scheme.
5     If no such marker exists, return 'None', showing that no scheme could be determined.
6     :return: string, If successful, return the scheme of the URL
7     """
```

Instruction 3

```
1 def get_scheme(self):
2     """
3     get the scheme of the URL
4     :return: string, If successful, return the scheme of the URL
5     >>> urlhandler = URLHandler("https://www.baidu.com/s?wd=aaa&rsv_spt=1#page")
6     >>> urlhandler.get_scheme()
7     "https"
8     """
```

38. In a few words, what is the task this function aims to implement? *

39. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

40. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

41. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 11:

Below are instructions for implementing the function

Instruction 1

```
1 def remove_duplicates(numbers: List[int]) -> List[int]:
2     """
3     Create a function called 'remove_duplicates' that accepts a list of integers 'numbers' as its parameter.
4     This function should return a new list containing only those integers which occur exactly once in the 'numbers'.
5     To do this, it first builds a dictionary 'num_cnt' to track how many times each integer appears.
6     For each integer, if it is not already present as a key in 'num_cnt', it initializes the count to 0
7     and then increases the count by 1 for each appearance.
8     Finally, the function returns a new list of integers using a list comprehension, including only those
9     integers for which their occurrence in 'num_cnt' is exactly 1, preserving their original order.
10    """
```

Instruction 2

```
1 def remove_duplicates(numbers: List[int]) -> List[int]:
2     """
3     Write a function 'remove_duplicates' that purges multiples
4     of any integer in a given list and outputs a new list consisting
5     solely of unique integers, in the same sequence as presented initially.
6     """
```

Instruction 3

```
1 def remove_duplicates(numbers: List[int]) -> List[int]:
2     """
3     From a list of integers, remove all elements that occur more than once.
4     Keep order of elements left the same as in the input.
5     >>> remove_duplicates([1, 2, 3, 2, 4])
6     [1, 3, 4]
7     """
```

42. In a few words, what is the task this function aims to implement? *

43. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

44. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

45. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Question 11:

Below are instructions for implementing the function:

Context

```
1 import math
2 class Statistics3:
3     """
4     This is a class that implements methods for calculating indicators such as median, mode,
5     correlation matrix, and Z-score in statistics.
6     """
7
8     def median(data):
9         pass
10
11     @staticmethod
12     def correlation(x, y):
13         pass
14
15     @staticmethod
16     def mean(data):
17         pass
18
19     @staticmethod
20     def correlation_matrix(data):
21         pass
22
23     @staticmethod
24     def standard_deviation(data):
25         pass
26
27     @staticmethod
28     def z_score(data):
29         pass
30
31
32
```


Instruction 1

```
1 @staticmethod
2 def mode(data):
3     """
4     Calculate the mode of the given list "data".
5     If multiple values have the same highest frequency,
6     return all of them in a list.
7     :param data: the given list, list.
8     :return: the mode of the given list, list.
9     """
```

Instruction 2

```
1 @staticmethod
2 def mode(data):
3     """
4     To calculate the mode from the list 'data', follow the steps outlined:
5     Utilize a dictionary 'counts' to record occurrences of each unique value in 'data',
6     enhancing each count as you go. Determine the utmost frequency and title this 'max_count'.
7     Then, create a list 'mode_values' composed of all values whose occurrence matches the 'max_count'.
8     Output the 'mode_values' list, making sure to include every value if multiple share the highest frequency.
9     :param data: the given list, list.
10    :return: the mode of the given list, list.
11    """
```

Instruction 3

```
1 @staticmethod
2 def mode(data):
3     """
4     calculates the mode of the given list.
5     :param data: the given list, list.
6     :return: the mode of the given list, list.
7     >>> statistics3 = Statistics3()
8     >>> statistics3.mode([1, 2, 3, 3])
9     [3]
10    """
```

46. In a few words, what is the task this function aims to implement? *

47. How would you characterize the subjective difficulty of the task? *

Une seule réponse possible.

- ☐ Very Easy
- ☐ Easy
- ☐ Not Easy Nor Difficult
- ☐ Difficult
- ☐ Very Difficult

48. How would you rate the difficulty of this task, in minutes it would take to implement? *

Une seule réponse possible.

- ☐ 1 min
- ☐ 5 min
- ☐ 10 min
- ☐ 20 min
- ☐ 40 min
- ☐ > 40 min

49. Why did you consider this task of such difficulty? Explain your reasoning (time it would take to search a concept, difficulty of the programming structure involved, etc.) *

Demographics Information

50. What is your highest education qualification? *

Une seule réponse possible.

- ☐ Ph.D.
- ☐ Master's Degree
- ☐ Bachelor Degree
- ☐ Autre :

51. What is your field of work? (Engineer, Researcher, Developer, etc.) *

52. How many years of development experience do you have? *

Une seule réponse possible.

- ☐ Less than 5
- ☐ Between 5 and 10
- ☐ More than 10

53. Thank you for completing the survey! Any feedback you would like to provide?

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